

Exam Date & Time: 18-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: MAT 2101/MAT_2101 DISCRETE MATHEMATICAL STRUCTURES

(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Discrete Mathematical Structures [MAT 2101]

Marks: 50

Duration: 180 mins.

A

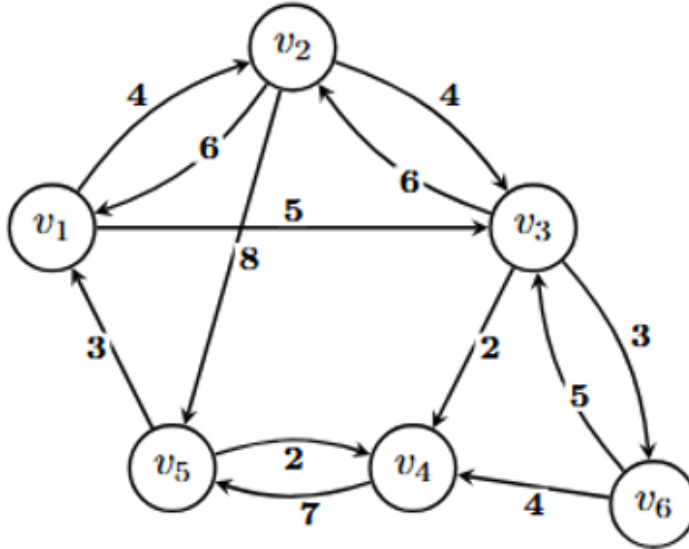
Answer all the questions.

- 1A) Draw the Hasse diagram for the Poset $A = \{1, 2, 3, 12, 18, 36\}$ with a partial ordering relation defined as: a is related to b if and only if a divides b . Is it a lattice? Justify your answer. Also list all the antichains of length more than one in the Poset A . (4)
- 1B) In a Boolean algebra $(A, \vee, \wedge, \neg)$, show that
 (i) every element has a unique complement.
 (ii) $a \vee (\bar{a} \wedge b) = a \vee b$ and $a \wedge (\bar{a} \vee b) = a \wedge b$ for all $a, b \in A$. (3)
- 1C) Let $E(x_1, x_2, x_3) = (\bar{x}_1 \wedge x_2 \wedge \bar{x}_3) \vee (x_1 \wedge \bar{x}_2) \vee (x_1 \wedge x_3)$ over $(\{0,1\}, \vee, \wedge, \neg)$ be a Boolean expression. Write $E(x_1, x_2, x_3)$ in both CNF and DNF. (3)
- 2A) How many ways are there to select a total of 15 items from a collection of three types of items where the selection of each type has the following conditions:
 i. Type A selection must be between 1 and 4 items.
 ii. Type B selection must be between 3 and 7 items.
 iii. Type C selection must be between 5 and 10 items. (4)
- 2B) Show that the number of partitions of n in which no part is smaller than 3, is equal to the number of partitions of $n+3$ in which the three largest parts are consecutive integers. (3)
- 2C) Find the 75th permutation of the five marks 0, 1, 2, 3, 4 in Fike's order. (3)
- 3A)
 (i) Prove that the set D of all odd integers forms an abelian group with respect to the binary operation $*$ defined by $a * b = a + b - 1 \forall a, b \in D$.
 (ii) Let H and K be subgroups of a group G with $HK = KH$. Then prove that HK is a subgroup of G . (4)
- 3B) Show that if G is a regular self-complementary graph of order n , then $n = 4t + 1$ for some integer t . (3)
- 3C) Solve
 $2x + y + z = 5$
 $3x + 5y + 2z = 15$
 $2x + y + 4z = 8$ by LU decomposition method. (3)

- 4A) Prove that i) Every cyclic group is abelian.
ii) Any group with at most five elements is abelian. (4)

- 4B) If G has p vertices and minimum degree $\delta(G) \geq \frac{p-1}{2}$, then prove that G is connected. (3)

- 4C) Use Dijkstra's algorithm to find the shortest paths from v_1 to all other vertices for the network given below. (3)



- 5A) Find a singular value decomposition of the matrix $\begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$. (4)

- 5B) Consider the following premises:
If there was a ball game, then travelling was difficult.
If they arrived on time, then travelling was not difficult.
They arrived on time.
Using propositional inference theory, show that there was no ball game. (3)

- 5C) Show that from the premises
 $(\exists y) (M(y) \wedge \neg W(y))$ and
 $(\exists x) (F(x) \wedge S(x)) \rightarrow (\forall y) (M(y) \rightarrow W(y))$
the conclusion $(\forall x) (F(x) \rightarrow \neg S(x))$ follows. (3)

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Event Name	Subject
058-END SEMSTER-ODD-2025	MAT_2125:ENGINEERING MATHEMATICS-III

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	9
4	4	CO 4	8

5	5	CO 5	6
6	6	CO 6	7
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	MAT_2125:ENGINEERING MATHEMATICS-III

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	11	36(72%)
4	Analysis	4	14(28%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)